Neutron Transport Through Homogeneous/Inhomogeneous Slab Material

1 Simulation

1.1 Physics and Model

In this simulation, we consider elastic and incoherent scattering of neutrons in a bulk material. For such a phenomenon, an incident neutron enters the material and will be prone to interact with the material, with what can be considered a fixed average rate β, which will be determined by the energy of the neutron.

These interactions consist of mainly two types; nuclear interaction, when the neutron interacts with the nuclei of the material, and magnetic interactions, when the magnetic field of the material atoms affect the neutron (having a non-zero dipole moment). As we will only look at the events of a neutron escaping the bulk or being absorbed, we neglect the magnetic interactions, as their effect is mainly to change the neutron trajectory by some small angle. As for the nuclear interaction, when the neutron interacts with a nuclei, one of two things will occur. The neutron may either be absorbed into the nuclei or be scattered into some random direction influenced by the incident direction and other factors at interaction. In the case the neutron is scattered, it may then travel further into the material until it again interacts with or escapes the bulk.

In between interactions, we model the neutron to traveling undisturbed as a particle. Normally, neutrons would also be able interact with each other during the transport process, causing what is called incoherent. But to investigate merely the penetrative ability of the neutron in a material, such interactions are negligible.

For this simulation, we assume a model where the nuclear interactions are equally likely to occur at any point in space traversed by the neutron. In reality, this would be dependent on the number density of the material, but for a reasonable length scale of radiation shielding, we can approximate the material to be quite dense. In such a case, the length l traveled by the neutron between interactions can be considered exponentially distributed, with a probability density

$$p_l(x) = \frac{1}{\lambda} e^{-x/\lambda} \tag{1}$$

for $x > 0$ and the mean free path λ, related to the interaction rate by $\lambda = v/\beta$ with neutron speed v.

At the event of interaction, the neutron is either absorbed or scattered. These events are random and are on average determined by the properties of the material in question. We therefore introduce parameters p_a and p_s as being the probabilities of absorption and scattering, respectively, also representing the

properties of the material in question. Although these two events in reality aren't completely exhaustive, e.g. there is a possibility of fission occurring in some materials for which the number of scattered neutrons increase, we do not include any other effects in our model. We therefore state in this model that $p_a + p_s = 1$. But we will allow to modify the definition of p_a and p_s as being position dependent in order to examine material inhomogeneity. In particular, we say that given that our material is a mix between two materials of different scattering properties, the values of p_a and p_s are set by which element nucleus the neutron interacts with. This will also be determined randomly based on the assumed proportion of material particles.

At scattering, we assume that the neutron is equally likely to scatter in any direction. This is not in general true, but should for most isolating materials be a good approximation. More formally, we say that the probability to scatter in the infinitesimal solid angle $\mathrm{d}\Omega = \sin\theta d\theta d\phi$, $p(\theta,\phi)d\theta d\phi$, is

$$p(\theta,\phi)d\theta d\phi = \frac{\sin\theta}{4\pi} d\theta d\phi \tag{2}$$

This also give probabilities for each angle

$$p_\theta = \int_0^{2\pi} p(\theta,\phi)d\phi = \frac{\sin\theta}{2}, \tag{3}$$

and

$$p_\phi = \int_0^{\pi} p(\theta,\phi)d\theta = \frac{1}{2\pi}. \tag{4}$$

In our simulation of a neutron trajectory, we make use of (1) and (2) to determine the trajectory after scattering events, although, in doing this, we make a simplification that the trajectory need only be observed in one dimension. In doing this, we assume the bulk material to have a depth d, say in the z-direction, being considerably smaller than the other dimension sizes for the length scale considered. Thus escape in any direction other than along the z-axis is negligible. We shall therefore only make use of (1) and (3) in calculating the distance traveled in the z-direction, $l\cos\theta$.

1.2 Monte Carlo simulation

Using the model above, we can sample neutron trajectories and calculate probabilities of incidents neutrons being reflected (exiting the material from incident side), absorbed, and/or transmitted (exiting the material on the other side), by calculating the fraction of simulated neutrons experiencing each case. In this simulation, picking a random angle and travel distance between interactions is needed. For that we pick a number k from a uniformly random number generator creating numbers between 0 and 1. Then for the random variable $X \in [a,b]$, with random distribution $p(x)$, we actually want to sample from, we define

$$k = \int_a^x p(s)ds = P(X < x) \tag{5}$$

which defines a corresponding random number $x \in [a,b]$ sampled from X. For the travel distance between interactions, this gives

$$l = -\lambda \ln k \tag{6}$$

and for the scattering direction, $\cos\theta$, gives

$$\cos\theta = 1 - 2k \tag{7}$$

For this simulation procedure, we consider a certain number of simulations, W, and do the following for each one:

1. Consider a neutron being incident to the material border plane, entering the material perpendicular to the plane, i.e. with $\cos\theta = 1$ and $z = 0$. We generate l by (6), and update the z position.

2. Now, the neutron is considered to interact with the material. A random number $k \in [0, 1]$ is drawn to determine if the neutron is absorbed ($k < p_a$) or scattered ($k \geq p_a$). If the material is a mix between two elements, of two different p_a with fraction s material 1, then with p_a is used in the previous procedure is randomized using s as the probability to interact with material 1 and $1 - s$ for material 2.

3. If it is absorbed, we increase the absorb counter $N_a = N_a + 1$ and repeat from step 1.

4. The neutron is scattered into an arbitrary direction, determining $\cos\theta$ according to (7), and travels a distance l determined by (6). The z position is updated according to $z_{\text{new}} = z + l\cos\theta$.

5. If the neutron has been reflected ($z < 0$), we update the reflection counter $N_c = N_c + 1$ and repeat from Step 1. If it has been transmitted through the material ($z > d$), then we update the transmission counter $N_t = N_t + 1$ and continue from Step 1. Otherwise, we repeat from Step 2.

A code listing for this simulation is given in the Appendix.

2 Problem

Using the model and Monte Carlo method described, we will investigate the behavior of neutron transmission through the slab when varying the different physical parameters, p_s, λ, d, involved. We will attempt to answer the question: How does the attenuation of the neutron transmission intensity change when varying the physical involved?

We will also investigate the change in the transport process when changing the absorption capabilities of the bulk material. In particular, we will examine how a uniform mix (material atoms uniformly randomly distributed in the bulk) between two materials when varying the proportions of the two material and compare with the case of a pure, homogeneous slab. We will attempt to answer the question: Could one increase the shielding capabilities of a slab better by proportions of a better shielding material, better than increase the depth of the slab?

3 Method

3.1 Measurement and analysis

In this Monte Carlo, we will measure the probabilities of transmission (p_t), reflection (p_r) and absorption (p_c) of neutrons as they are affected by the parameters involved. Any probability p_i which we attempt to determine, will be measured as the fraction of neutrons experiencing the given event out of the W simulations performed, i.e. we measure

$$p_i = \frac{N_i}{W} = \frac{1}{W}\sum_{k=1}^{W} x_{i,k} \tag{8}$$

using the outcomes of the stochastic variables $X_{i,k} \in \{0, 1\}$ describing measurement of the kth neutron experiencing the specified event i.

To obtain error estimates of obtained values, we make use of the Central Limit Theorem. For large enough W, which limit we will have to examine, the mean values of stochastic variable outcomes, $x_{i,k}$, will be approximately normally distributed, and the standard deviation of the mean value calculated from (8) can be approximated by

$$\sigma_i^2 = \frac{\sum_{k=1}^{W}(x_{i,k} - p_i)^2}{W^2} \tag{9}$$

for large enough W. Knowing the variance of the normally distributed mean values, we can construct a 99% confidence interval

$$I_{\text{conf},i} = [p_i - 2.576\sigma_i, p_i + 2.576\sigma_i] \tag{10}$$

having 99% probability of enclosing the correct value. The size of this interval is used as our error estimate.

But other than just using these error measurements, we need to determine the number W required to obtain a small enough error, while also analyzing the sensitivity of the error as we vary our parameters. As such, for the first basic test of the model, we vary the number of iterations and observe how the error changes.

3.2 Attenuation

First we perform a test to see how well our model correspond to known behavior of neutron radiation, by measuring the attenuation of the transmitted neutron intensity (corresponding to the transmission probability p_t) as a function of neutron energy and depth of the isolating bulk.

The Beer-Lambert law of exponential attenuation is known to hold for neutron radiation, so we should expect exponential reduction of transmitted intensity when increasing bulk depth.

The increase of neutron energy in our model corresponds to an increase in the mean free path λ. Also, since an increase in λ is the same as reducing the average number of nuclear interactions needed before being transmitted, one can suspect that increasing λ would have the same effect as reducing the depth d. We therefore also examine the transmission intensity as a function of λ and the possible similarity to the dependence on d.

3.3 Material inhomogeneity

We examine the dependence of event probabilities on the material properties of scattering and absorption probabilities. In an attempt to answer how material impurities can affect the system, we let the bulk be a mix between materials of varying absorption capabilities, and vary the proportion between materials. We compare the probabilities obtained from mixing two materials with the ones obtained from letting the bulk consist entirely of one material.

4 Results and Analysis

4.1 Attenuation

In fig. 1 we measured the transmission probability for different bulk depths, using $\lambda = 1$, $W = 10^5$ and $p_s = \frac{2}{3}$. The estimated error, calculated according to (10), is also displayed at the data points.

Although, as can be seen in the figure and fig. 4 shows, the error estimates obtained are less than 0.005 for these parameter configurations. This continues to be true for the results obtained in this section. An exponential function on the form of ae^{bx} was fitted to the data in order to test if the Beer-Lambert law hold for this model.

In fig. 2 we measured the transmission probability using the same parameter configuration as for fig. 1, but for varying mean free path of the neutrons, instead of bulk depth. For this data, we also fit a function on the form $ae^{b/x}$, from the intuition that increasing mean free path reduces the average number of scattering/absorption events the same way as reducing the depth would.

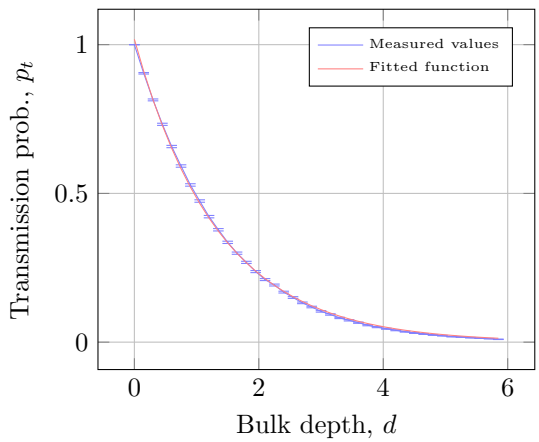

Figure 1: Transmission probability as function of the bulk depth d, measured using $W = 10^5$, $\lambda = 1$ and $p_s = \frac{2}{3}$. Also showing fitted exponential function $y = ae^{bx}$.

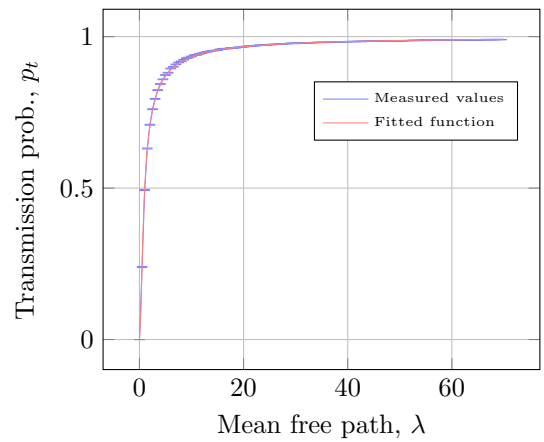

Figure 2: Transmission probability as function of the mean free path λ, measured using $W = 10^5$, $d = 1$ and $p_s = \frac{2}{3}$. Also showing fitted reciprocal exponential function $y = ae^{b/x}$.

As the data from both figs. 1 and 2 show good agreement with assumed exponential decays, we test the attenuation law for the combination of the two paramters, formed as the ratio $r = d/\lambda$. Keeping the same configuration as before, but varying the ratio r, choosing $t = \sqrt{r}$ and $\lambda = 1/\sqrt{r}$, we obtain fig. 3.

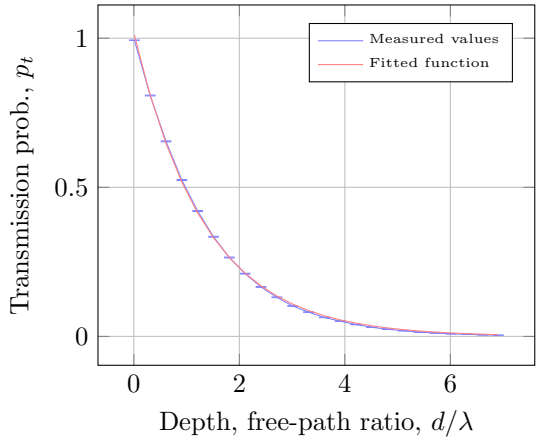

Figure 3: Transmission probability as function of the time and free path ratio r, measured using $W = 10^5$, $p_s = \frac{2}{3}$ and $d = \sqrt{r}$, $\lambda = 1/\sqrt{r}$, such that $r = d/\lambda$. Also showing fitted function $y = ae^{br}$.

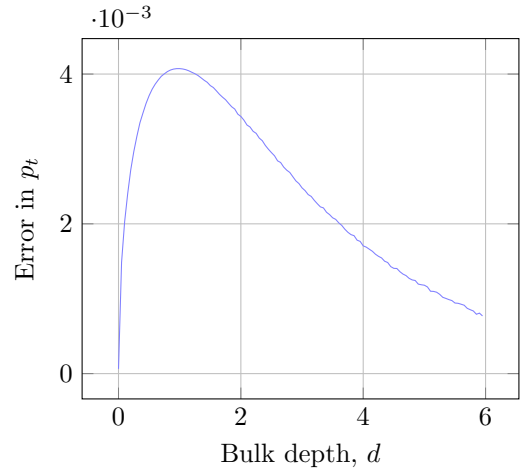

Figure 4: Calculated error estimate as function of d obtained from test in fig. 1. Similar trends obtained for rest of attenuation test data.

We test for convergence of results for other parameter configurations of W and p_s in order to make sure that our obtained results are consistent. We do this, first by running the same simulation for different number of samples W, for different depths d, keeping $\lambda = 1$ and $p_s = \frac{2}{3}$ constant. The convergence plots in fig. 5 show measured error estimate as a function of W for different d and table 1 show the resulting transmission probability and error measured at the most exact measurement for each case of d.

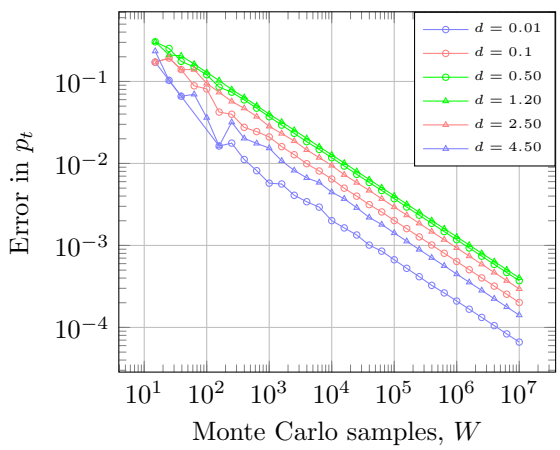

Figure 5: Error measured as function of number of simulations W, for parametric setups keeping $\lambda = 1$ and $p_s = \frac{2}{3}$ while varying d.

d	p_t at $W = 10^7$	p_t error
0.01	0.99334	$\pm 6.6 \cdot 10^{-5}$
0.10	0.9345	$\pm 2.0 \cdot 10^{-4}$
0.50	0.7064	$\pm 3.7 \cdot 10^{-4}$
1.20	0.4234	$\pm 4.0 \cdot 10^{-4}$
2.50	0.1556	$\pm 3.0 \cdot 10^{-4}$
4.50	0.0311	$\pm 1.4 \cdot 10^{-4}$

Table 1: Table of d values used for convergence tests in fig. 5, and corresponding value of transmission probability p_t obtained at the highest value of W tested.

From fig. 1 and **??**, we saw that the error reaches a maximum for probabilities around $p_t \approx 0.5$, and from fig. 6 and table 1, we draw the same conclusion, also noting that the error is smaller than 0.01 for

all d if measurements are performed with $W > 10^5$ samples. We therefore continue using $W = 10^5$ for further simulations.

Finally, we test if the exponential behavior of the attenuation is affected by the scattering probability of the material. We redo the test done in fig. 1 using different scattering probabilities p_s and fit exponential functions to the data in each case. fig. 6 shows the obtained data plots and table 2 contains obtained parameter values as well as the R-squared goodness of fit measure.

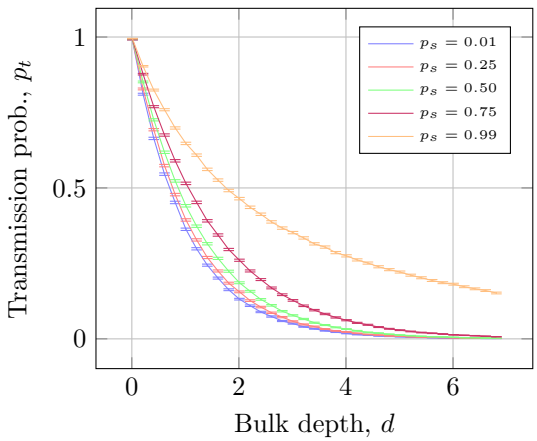

Figure 6: Dependence of transmission probability on d as done in fig. 1, remeasured for different probabilities of scattering p_s.

p_s	b	R^2
0.01	-0.997	0.99998
0.25	-0.932	0.99993
0.50	-0.836	0.99974
0.75	-0.682	0.99968
0.99	-0.301	0.9841

Table 2: Exponential coefficient b in fit $y = ae^{bx}$ for data in fig. 6, for the different p_s cases. Goodness of fit measure R^2 included.

4.2 Material Inhomogeneity

Here we wish to investigate how the overall absorption probability p_c of the procedure is affected by material properties. We first measure the absorption probability of pure bulk materials of different internal scattering probabilities p_s. We use a parameter configuration to obtain a reasonable maximum probability and error when using medium p_s as in earlier studies, i.e. $\lambda = d = 1$ and $W = 10^5$. The result of this is shown in fig. 7.

Next we introduce impurities of varying degree into the material. We assume that original material being mixed has a scattering probability $p_s = 0.2$ (p_a defined through $p_a = 1 - p_s$) and is mixed with a material with $p_s = 0.8$. Simulating with the same parameter configuration as in fig. 7 and using the percentage of $p_a = 0.8$ material atoms as a measure, we obtain fig. 8. Which shows a more linear transition of absorption probabilities from that of the $p_s = 0.8$ material, to that of the $p_s = 0.2$ material, when comparing to fig. 7.

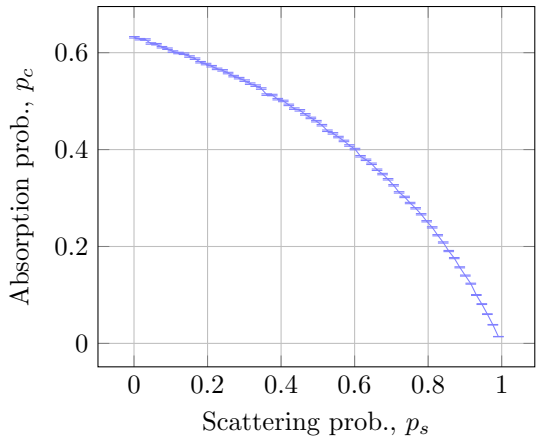

Figure 7: Probability of absorption as function of scattering probability p_s, for $d = \lambda = 1$ and $W = 10^5$, letting bulk be homogeneous.

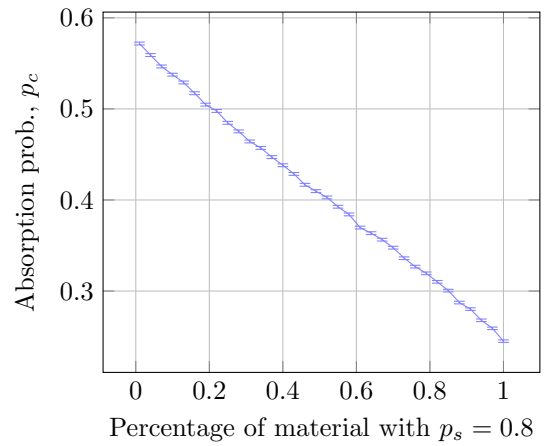

Figure 8: Probability of absorption as function of the percentage of $p_s = 0.8$ material, then letting $1 - x$ be the percentage of $p_s = 0.2$ material.

5 Discussion and Conclusions

In the Attenuation section, we the fitted exponential function in fig. 1 does not deviate from the data in any aspect measurable in the figure and is enclosed by the error bounds of the data points. We can therefore conclude that for the parameter configuration tested, the attenuation follows Beer-Lambert's law, being exponentially decreasing with slab depth.

As we further tested the behavior when instead varying the mean free path of the neutrons, the fitted exponential function $ae^{b/x}$ fit the data just as well as in the last case, with measurements having a negligible error of less than 0.01 for all λ tested. When varying ratio instead of each parameter separately in fig. 3, we had errors just as small as the ones measured in the previous, and there too an exponential function fit to the data, with no mentionable deviation from the data. Considering the small errors between fitted function and data points, we can safely conclude that the exponential attenuation holds for this model for the investigated parameter configurations, as it does with real neutron transmission. This, despite neglecting not obviously small contributions such as fission during nuclear interaction.

Performing the convergence test in fig. 5 and examining the measured error during the tests, such as in fig. 4, we see that the error varies with the measured result. The error seems maximum for results measuring $p \approx 0.5$ and is reduced as the distance to either $p = 0$ or $p = 1$ is. This is likely an effect of averaging results of a discrete stochastic variable only taking values of 0 and 1. As the values of the mean are bounded below by 0 and above 1, the variance can be predicted to be reduced for mean values measured close to 0 or 1. In the $p \approx 0.5$ case, the mean values can vary maximally in both directions, causing a greater variance to be measured.

Further examining the behavior of the error estimate at low probability results, we see that while the error is reduced, the number of Monte Carlo samples W needed to converge toward the reduction trend is larger than for the other measurements. Both this and the reduced error for these measurements could come from the fact that we have a certain uncertainty in our result not completely encapsulated by the error estimation done according to (10). Since we are measuring probabilities as fractions of scores kept

by repeated simulation, in order to measure a probability of order 10^{-k}, one would need a number of measurements W of the order 10^k or more. Thus it is not odd that the estimated error fluctuates more for $p \approx 0.01$ when we are only using between $W = 10^2$ and $W = 10^3$ measurements. It is important to keep this in mind when using this kind of Monte Carlo simulation, relying on evaluation of discrete stochastic variables.

But all in all, judging from the convergence plot in fig. 5, the error seems to decrease asymptotically according $O(W^{-1/2})$, and all measurements reach a satisfactory and probable error of less than 10^{-2} for $W = 10^5$, which was used in the rest of our measurements.

After taking this error analysis into consideration, the continued investigation of attenuation, now analyzing the effect of the scattering probability p_s showed in fig. 6 and table 2 that decreasing p_s changes the exponent. Fitted exponential functions still had extremely good agreement with obtained data, with the only case standing out was for $p_s = 0.99$ with a goodness of fit measure $R^2 \approx 0.98$. This shows that the exponential attenuation law shown to hold for one parameter configuration, also hold when changing other physical parameters. To answer the question we stated in the problem statement: The attenuation of transmission intensity during neutron transport through a slab is exponential in the ratio of depth and mean free path, with the rate of intensity decay determined by the scattering probability of the material.

Regarding the absorption probability p_c as a function of the scattering probability of the slab, creating a mixture of two materials seems to create a material with average absorbing capabilities being the mean values of the materials, weighted by the respective fraction of material atoms, as indicated by the linear trend in fig. 8. This means that by adding better absorbents to the slab, it would increase the material's overall absorbing ability, but one would achieve a greater improvement by replacing the entire slab by a slightly better absorbent. Since increasing the depth also would increase absorptions considerably (as p_t decreases exponentially and total probability is conserved $p_t + p_c + p_r = 1$), creating mixed materials would probably not compare. The accuracy of this result though, is debatable, since it is based on the mixing of materials being uniform. The considerable simplification in our model of randomly picking the interacting material would not hold for slightly non-uniform material distributions and also does not account for neutrons in the simulations arriving in the same location twice. One should probably have to add certain details to the model such as defining intervals for the nuclei positions if keeping a discrete model is desirable. Although this would also mean a large amount of memory overhead for any decent accuracy, and is thus not attempted here.

6 Appendix: Code listings

Code listing showing C++ code for neutron transport simulating neutrons traveling through slabs with constant probabilities of scattering p_s and absorption $p_a = 1 - p_s$, with physical parameters, slab depth d and mean free path λ, and a number of simulations W.

```
input: lambda, d, p_s, p_a and W
enum Cases {absorb=0, reflect, scatter};

seed = chrono::system_clock::now().time_since_epoch().count();
mt19937 gen(seed);
uniform_real_distribution<double> dist(0, 1); // number generator
vector<int> result(3, 0);
vector<vector<int>> vals(W, vector<int>(3, 0));
```

```
double z = 0;
double rand_val = 0;
for (int l = 0; l < W; l++)
{
    string input;
    z = 0;
    while (z >= 0 && z <= d)
    {
        if (z != 0) // if this isn't starting point
        {
            rand_val = dist(gen);
            if (rand_val <= p_a) // absorption
                break;
        }

        // random angle and travel distance
        double angle_rand = z == 0 ? 0 : dist(gen);
        double length_rand = dist(gen);
        z += -lambda*log(length_rand)*(1-2*angle_rand);
    }

    // save result
    if (z < 0)
    {
        result[reflect]++;
        history[l][reflect]++;
    }
    else if (z > d)
    {
        result[scatter]++;
        history[l][scatter]++;
    }
    else
    {
       result[absorb]++;
       history[l][absorb]++;
    }
}

// calculation of mean and variance using result and vals...
```